# 第9章　一元函数积分学的计算

**1.【答案】**1

**【解析】**令 $t=\ln\dfrac{1}{1-x}$，则 $x=1-\mathrm{e}^{-t}$，故

$$原式=\int_0^{+\infty}t\mathrm{d}(1-\mathrm{e}^{-t})=\int_0^{+\infty}t\cdot\mathrm{e}^{-t}\mathrm{d}t=1.$$

**2.【解析】**设 $\dfrac{x+2}{(2x+1)(x^2+x+1)}=\dfrac{A}{2x+1}+\dfrac{Bx+D}{x^2+x+1}$，通分后可得

$$x+2=A(x^2+x+1)+(Bx+D)(2x+1),$$

代入特值 $x=-\dfrac{1}{2}$，$x=0$，$x=1$，得 $A=2$，$D=0$，$B=-1$，故

$$原式=\int\left(\frac{2}{2x+1}-\frac{x}{x^2+x+1}\right)\mathrm{d}x=\ln|2x+1|-\int\frac{x}{x^2+x+1}\mathrm{d}x$$

$$=\ln|2x+1|-\frac{1}{2}\int\frac{(2x+1)-1}{x^2+x+1}\mathrm{d}x$$

$$=\ln|2x+1|-\frac{1}{2}\int\frac{\mathrm{d}(x^2+x+1)}{x^2+x+1}+\frac{1}{2}\int\frac{\mathrm{d}x}{\left(x+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}$$

$$=\ln|2x+1|-\frac{1}{2}\ln(x^2+x+1)+\frac{1}{\sqrt{3}}\arctan\frac{2x+1}{\sqrt{3}}+C.$$

**3.【答案】**$\dfrac{1}{2}\mathrm{e}(\cos 1+\sin 1)-\dfrac{1}{2}$

**【解析】**令 $\ln x=t$，则 $x=\mathrm{e}^t$，故

$$原式=\int_0^1\mathrm{e}^t\cos t\mathrm{d}t=\frac{1}{2}(\mathrm{e}^t\cos t+\mathrm{e}^t\sin t)\Big|_0^1=\frac{1}{2}\mathrm{e}(\cos 1+\sin 1)-\frac{1}{2}.$$

**4.【解析】**以 $1-x$ 代换原方程中的 $x$，得 $(1-x)f(1-x)+f(x)=(1-x)^2$. 此方程与原方程联立，解得 $f(x)=\dfrac{x^3-2x+1}{x^2-x+1}$. 于是

$$\int f(x)\mathrm{d}x=\int\frac{x^3-2x+1}{x^2-x+1}\mathrm{d}x=\int\left(x+1+\frac{-2x}{x^2-x+1}\right)\mathrm{d}x$$

配套课程公众号[研大]
联系微信19316378888

$$=\frac{1}{2}x^2+x+\int\frac{-(2x-1)-1}{x^2-x+1}\mathrm{d}x$$

$$=\frac{1}{2}x^2+x-\int\frac{\mathrm{d}(x^2-x+1)}{x^2-x+1}-\int\frac{\mathrm{d}\left(x-\frac{1}{2}\right)}{\left(x-\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}$$

$$=\frac{1}{2}x^2+x-\ln(x^2-x+1)-\frac{2}{\sqrt{3}}\arctan\frac{2x-1}{\sqrt{3}}+C.$$

**5.【答案】** $\frac{1}{\ln 2}+\frac{1}{\ln^2 2}$

**【解析】** $$\text{原式}=\int_1^{+\infty}2^{-\sqrt{x}}\mathrm{d}x\overset{\sqrt{x}=t}{=\!=\!=}\int_1^{+\infty}2^{-t}\cdot 2t\mathrm{d}t$$

$$=\frac{-2}{\ln 2}\int_1^{+\infty}t\mathrm{d}(2^{-t})=-\frac{2}{\ln 2}\left(t\cdot 2^{-t}\Big|_1^{+\infty}-\int_1^{+\infty}2^{-t}\mathrm{d}t\right)$$

$$=-\frac{2}{\ln 2}\left(-\frac{1}{2}+\frac{2^{-t}}{\ln 2}\Big|_1^{+\infty}\right)=\frac{1}{\ln 2}+\frac{1}{\ln^2 2}.$$

**6.【答案】**4

**【解析】**令 $x=1+t$，由 $f(x)$ 是连续的偶函数，且 $\int_0^1 f(x)\mathrm{d}x=2$，得

$$\int_0^2 xf(1-x)\mathrm{d}x=\int_{-1}^1(1+t)f(-t)\mathrm{d}t=2\int_0^1 f(t)\mathrm{d}t=4.$$

**7.【答案】** $\frac{5}{3}$

**【解析】**令 $F(x)=\int_{x^2}^{x^2+1}f(t)\mathrm{d}t(x>0)$，则 $F'(x)=2x\left[f(x^2+1)-f(x^2)\right]=2x^2$，则

$$\int_{x^2}^{x^2+1}f(t)\mathrm{d}t=F(x)=\frac{2}{3}x^3+C.$$

取极限 $x\to 0^+$，得 $\int_0^1 f(t)\mathrm{d}t=C$，即 $C=1$，则 $F(x)=\frac{2}{3}x^3+1(x>0)$，故

$$\int_1^2 f(x)\mathrm{d}x=F(1)=\frac{2}{3}+1=\frac{5}{3}.$$

**8.【解析】**由 $f(t)=\int_0^1 t|t-x|\,\mathrm{d}x\overset{t-x=u}{=\!=\!=}\int_t^{t-1}t|u|(-\mathrm{d}u)=t\int_{t-1}^t|u|\,\mathrm{d}u$，故

当 $t\geqslant 1$ 时，$f(t)=t\cdot\frac{1}{2}u^2\Big|_{t-1}^t=t\cdot\left(t-\frac{1}{2}\right)=t^2-\frac{t}{2}$；

配套课程公众号[研大]
联系微信19316378888

当 $0<t<1$ 时，$f(t)=t\cdot\frac{1}{2}u^2\Big|_0^{t-1}+t\cdot\frac{1}{2}u^2\Big|_0^{t}=t^3-t^2+\frac{t}{2}$；

当 $t\leqslant 0$ 时，$f(t)=t\cdot\frac{1}{2}u^2\Big|_t^{t-1}=-t^2+\frac{t}{2}$.

所以有

$$\begin{aligned}\int_{-1}^{2}f(t)\mathrm{d}t&=\int_{-1}^{0}\left(\frac{t}{2}-t^2\right)\mathrm{d}t+\int_{0}^{1}\left(t^3-t^2+\frac{t}{2}\right)\mathrm{d}t+\int_{1}^{2}\left(t^2-\frac{t}{2}\right)\mathrm{d}t\\&=\left(\frac{1}{4}t^2-\frac{1}{3}t^3\right)\Big|_{-1}^{0}+\left(\frac{1}{4}t^4-\frac{1}{3}t^3+\frac{1}{4}t^2\right)\Big|_{0}^{1}+\left(\frac{1}{3}t^3-\frac{1}{4}t^2\right)\Big|_{1}^{2}\\&=-\frac{7}{12}+\frac{1}{6}+\frac{19}{12}=\frac{7}{6}.\end{aligned}$$

**9.【答案】**(C)

**【解析】**由 $g(x)$ 是过点 $\left(-\frac{1}{2},0\right)$ 和 $(0,1)$ 的直线，可得 $g(x)=2x+1$，故

$$\begin{aligned}\int_0^2 f\left[g(x)\right]\mathrm{d}x&=\int_0^2 f(2x+1)\mathrm{d}x\xlongequal{2x+1=t}\int_1^5 f(t)\mathrm{d}\left(\frac{t-1}{2}\right)\\&=\frac{1}{2}\int_1^5 f(t)\mathrm{d}t=\frac{1}{2}\cdot 2\int_0^2 f(t)\mathrm{d}t=1.\end{aligned}$$

**10.【答案】**$\frac{\pi}{8}-\frac{1}{4}\ln 2$

**【解析】**$$f(x)=\int_0^x f'(t)\mathrm{d}t+f(0)=\int_0^x\arctan(t-1)^2\mathrm{d}t,$$

$$\begin{aligned}\int_0^1 f(x)\mathrm{d}x&=\int_0^1\mathrm{d}x\int_0^x\arctan(t-1)^2\mathrm{d}t\xlongequal{\text{交换次序}}\int_0^1\mathrm{d}t\int_t^1\arctan(t-1)^2\mathrm{d}x\\&=\int_0^1(1-t)\arctan(t-1)^2\mathrm{d}t\xlongequal{(1-t)^2=u}\frac{1}{2}\int_0^1\arctan u\mathrm{d}u=\frac{\pi}{8}-\frac{1}{4}\ln 2.\end{aligned}$$

**11.【答案】**$\frac{\pi}{6}$

**【解析】**由题意，$f^2(x)=-x^2+2x+3$，且 $f(x)>0$，所以

$$f(x)=\sqrt{-x^2+2x+3}=\sqrt{4-(x-1)^2},$$

$$\int_0^1\frac{1}{f(x)}\mathrm{d}x=\int_0^1\frac{1}{\sqrt{4-(x-1)^2}}\mathrm{d}x\xlongequal{x-1=t}\int_{-1}^0\frac{1}{\sqrt{2^2-t^2}}\mathrm{d}t=\arcsin\frac{t}{2}\Big|_{-1}^{0}=\frac{\pi}{6}.$$

配套课程公众号[研大]
联系微信19316378888

**12.【解析】** $\int_{-1}^{1}x\ln(1+e^{x})dx=\int_{-1}^{0}x\ln(1+e^{x})dx+\int_{0}^{1}x\ln(1+e^{x})dx$

$$=\int_{1}^{0}(-x)\ln(1+e^{-x})(-dx)+\int_{0}^{1}x\ln(1+e^{x})dx$$

$$=-\int_{0}^{1}x\ln(1+e^{-x})dx+\int_{0}^{1}x\ln(1+e^{x})dx$$

$$=\int_{0}^{1}x\ln\frac{1+e^{x}}{1+e^{-x}}dx=\int_{0}^{1}x\ln e^{x}dx=\int_{0}^{1}x^{2}dx=\frac{1}{3}.$$

**【注】原函数不易求的非奇非偶函数在对称区间上的积分，可采用此题方法处理.**

**13.【解析】** 注意到 $\frac{1}{(x+1)(x^{2}+1)}=\frac{1}{2(x+1)}-\frac{x-1}{2(x^{2}+1)}$，故

$$\int_{0}^{1}\frac{dx}{(x+1)(x^{2}+1)}=\frac{1}{2}\int_{0}^{1}\frac{dx}{x+1}-\frac{1}{4}\int_{0}^{1}\frac{d(x^{2}+1)}{x^{2}+1}+\frac{1}{2}\int_{0}^{1}\frac{dx}{x^{2}+1}$$

$$=\frac{1}{2}\ln(x+1)\Big|_{0}^{1}-\frac{1}{4}\ln(x^{2}+1)\Big|_{0}^{1}+\frac{1}{2}\arctan x\Big|_{0}^{1}=\frac{\pi}{8}+\frac{1}{4}\ln 2.$$

**14.【解析】** 分离变量得 $F(x)\mathrm{d}\left[F(x)\right]=\frac{\mathrm{d}(\sin x)}{1+\sin^{2}x}$，故 $F^{2}(x)=2\arctan(\sin x)+C$. 将 $F(0)=\sqrt{\pi}$ 代入得 $C=\pi$，又 $F(x)>0$，故 $F(x)=\sqrt{2\arctan(\sin x)+\pi}$.

**15.【解析】** $\int_{0}^{1}\frac{\arcsin\sqrt{x}}{\sqrt{x(1-x)}}dx=2\int_{0}^{1}\frac{\arcsin\sqrt{x}}{\sqrt{1-x}}\mathrm{d}(\sqrt{x})=2\int_{0}^{1}\arcsin\sqrt{x}\mathrm{d}(\arcsin\sqrt{x})=\frac{\pi^{2}}{4}$.

**16.【解析】** $\int_{0}^{\frac{\pi}{4}}\frac{dx}{\sin^{2}x+3\cos^{2}x}=\int_{0}^{\frac{\pi}{4}}\frac{dx}{\cos^{2}x(3+\tan^{2}x)}\xlongequal{t=\tan x}\int_{0}^{1}\frac{dt}{3+t^{2}}$

$$=\frac{1}{\sqrt{3}}\arctan\frac{t}{\sqrt{3}}\Big|_{0}^{1}=\frac{\pi}{6\sqrt{3}}.$$

**17.【解析】** $x=2$ 为奇点，应将积分区间分为两部分，即

$$\int_{1}^{3}\frac{dx}{\sqrt{|2x-x^{2}|}}=\int_{1}^{2}\frac{dx}{\sqrt{2x-x^{2}}}+\int_{2}^{3}\frac{dx}{\sqrt{x^{2}-2x}}$$

$$=\int_{1}^{2}\frac{dx}{\sqrt{1-(x-1)^{2}}}+\int_{2}^{3}\frac{dx}{\sqrt{(x-1)^{2}-1}}$$

$$=\arcsin(x-1)\Big|_{1}^{2}+\ln\left[x-1+\sqrt{(x-1)^{2}-1}\right]\Big|_{2}^{3}=\frac{\pi}{2}+\ln(2+\sqrt{3}).$$

配套课程公众号[研大]
联系微信19316378888

**18.【答案】** $\frac{(-1)^n}{3^{n+1}}n!$

**【解析】** 记
$$a_n=\int_0^1 x^2\ln^n x\mathrm{d}x=\int_0^1 \ln^n x\mathrm{d}\left(\frac{1}{3}x^3\right)$$
$$=\frac{1}{3}x^3\ln^n x\Big|_0^1-\int_0^1\frac{1}{3}x^3\cdot n\ln^{n-1}x\cdot\frac{1}{x}\mathrm{d}x$$
$$=-\frac{n}{3}\int_0^1 x^2\ln^{n-1}x\mathrm{d}x=-\frac{n}{3}a_{n-1},\ n=1,2,\cdots,$$

于是
$$a_n=-\frac{n}{3}a_{n-1}=\left(-\frac{n}{3}\right)\left(-\frac{n-1}{3}\right)a_{n-2}=\cdots=\left(-\frac{n}{3}\right)\left(-\frac{n-1}{3}\right)\cdots\left(-\frac{1}{3}\right)a_0,$$

又 $a_0=\int_0^1 x^2\mathrm{d}x=\frac{1}{3}$，故 $a_n=\frac{(-1)^n}{3^{n+1}}n!$.

**19.【答案】** (B)

**【解析】** 当 $0<x\leqslant 1$ 时，
$$f(x)=y\ln\sqrt{x^2+y^2}\Big|_{y=0}^{y=1}-\int_0^1 y\cdot\frac{1}{\sqrt{x^2+y^2}}\cdot\frac{2y}{2\sqrt{x^2+y^2}}\mathrm{d}y$$
$$=\ln\sqrt{x^2+1}-\int_0^1\frac{y^2}{x^2+y^2}\mathrm{d}y=\ln\sqrt{x^2+1}-\int_0^1\frac{x^2+y^2-x^2}{x^2+y^2}\mathrm{d}y$$
$$=\ln\sqrt{x^2+1}-1+\int_0^1\frac{x^2}{x^2+y^2}\mathrm{d}y=\ln\sqrt{x^2+1}-1+x\int_0^1\frac{1}{1+\left(\frac{y}{x}\right)^2}\mathrm{d}\left(\frac{y}{x}\right)$$
$$=\ln\sqrt{x^2+1}-1+x\arctan\frac{1}{x},$$

又 $f(0)=\int_0^1\ln y\mathrm{d}y=-1$，故
$$f'_+(0)=\lim_{x\to0^+}\frac{f(x)-f(0)}{x-0}=\lim_{x\to0^+}\frac{\ln\sqrt{x^2+1}-1+x\arctan\frac{1}{x}+1}{x}$$
$$=\lim_{x\to0^+}\frac{\ln\sqrt{x^2+1}}{x}+\lim_{x\to0^+}\arctan\frac{1}{x}$$
$$=\lim_{x\to0^+}\frac{\sqrt{x^2+1}-1}{x}+\frac{\pi}{2}=0+\frac{\pi}{2}=\frac{\pi}{2}.$$

**20.【解析】** 由题设知，积分变量 $t$ 与求导变量 $x$ 的取值在同一区间，不需要分情况讨论，故
$$I(x)=\int_{-1}^1|t-x|\mathrm{e}^{2t}\mathrm{d}t=\int_{-1}^x(x-t)\mathrm{e}^{2t}\mathrm{d}t+\int_x^1(t-x)\mathrm{e}^{2t}\mathrm{d}t$$

配套课程公众号[研大]
联系微信19316378888

$$=x\int_{-1}^{x}e^{2t}dt-\int_{-1}^{x}te^{2t}dt+\int_{x}^{1}te^{2t}dt-x\int_{x}^{1}e^{2t}dt,$$

$$I'(x)=\int_{-1}^{x}e^{2t}dt+xe^{2x}-xe^{2x}-xe^{2x}-\int_{x}^{1}e^{2t}dt+xe^{2x}=\int_{-1}^{x}e^{2t}dt-\int_{x}^{1}e^{2t}dt$$

$$=e^{2x}-\frac{1}{2}(e^{2}+e^{-2})\overset{令}{=\!=}0,$$

得 $x_0=\frac{1}{2}\ln\frac{e^2+e^{-2}}{2}$ 为唯一驻点，$I''(x)=2e^{2x}>0$，故 $x_0=\frac{1}{2}\ln\frac{e^2+e^{-2}}{2}$ 为 $I(x)$ 在 $[-1,1]$ 上的最小值点，最大值只能在端点 $x=-1$，$x=1$ 处取得．又

$$I(-1)=\frac{3}{4}e^{2}+\frac{1}{4}e^{-2},\ I(1)=\frac{1}{4}e^{2}-\frac{5}{4}e^{-2},$$

所以 $[I(x)]_{\max}=I(-1)=\frac{3}{4}e^{2}+\frac{1}{4}e^{-2}$．

**21.【解析】**由题设知，积分变量 $t$ 与求导变量 $x$ 的取值不在同一区间，需要分情况讨论．

当 $0<x\leqslant 1$ 时，
$$f(x)=\int_{0}^{x}\left|t^2-x^2\right|dt+\int_{x}^{1}\left|t^2-x^2\right|dt$$

$$=\int_{0}^{x}(x^2-t^2)dt+\int_{x}^{1}(t^2-x^2)dt=\frac{4}{3}x^3-x^2+\frac{1}{3};$$

当 $x>1$ 时，
$$f(x)=\int_{0}^{1}(x^2-t^2)dt=x^2-\frac{1}{3}.$$

所以
$$f(x)=\begin{cases}\frac{4}{3}x^3-x^2+\frac{1}{3}, & 0<x\leqslant 1,\\ x^2-\frac{1}{3}, & x>1,\end{cases}$$

而
$$f'_{-}(1)=\lim_{x\to 1^-}\frac{\frac{4}{3}x^3-x^2+\frac{1}{3}-\frac{2}{3}}{x-1}=2,\ f'_{+}(1)=\lim_{x\to 1^+}\frac{x^2-\frac{1}{3}-\frac{2}{3}}{x-1}=2,$$

故
$$f'(x)=\begin{cases}4x^2-2x, & 0<x\leqslant 1,\\ 2x, & x>1.\end{cases}$$

由 $f'(x)=0$ 得唯一驻点 $x=\frac{1}{2}$，又 $f''\left(\frac{1}{2}\right)>0$，从而 $x=\frac{1}{2}$ 为 $f(x)$ 的极小值点，即为最小值点，最小值为 $f\left(\frac{1}{2}\right)=\frac{1}{4}$．

**22.【解析】**
$$F(x)=\int_{0}^{x}f(t)g(x-t)dt\overset{x-t=u}{=\!=}\int_{0}^{x}f(x-u)g(u)du$$

$$=x\int_{0}^{x}g(u)du-\int_{0}^{x}ug(u)du.$$

配套课程公众号[研大]
联系微信19316378888

当$0\leqslant x\leqslant\pi$时，

$$F(x)=x\int_0^x\cos u\mathrm{d}u-\int_0^x u\cos u\mathrm{d}u=x\sin x-\int_0^x u\mathrm{d}(\sin u)=1-\cos x.$$

当$x>\pi$时，

$$F(x)=x\left[\int_0^\pi g(u)\mathrm{d}u+\int_\pi^x g(u)\mathrm{d}u\right]-\left[\int_0^\pi ug(u)\mathrm{d}u+\int_\pi^x ug(u)\mathrm{d}u\right]$$

$$=x\left(\int_0^\pi\cos u\mathrm{d}u+\int_\pi^x 0\mathrm{d}u\right)-\left(\int_0^\pi u\cos u\mathrm{d}u+\int_\pi^x 0\mathrm{d}u\right)=2.$$

于是，$F(x)=\begin{cases}1-\cos x, & 0\leqslant x\leqslant\pi,\\ 2, & x>\pi.\end{cases}$

**23.【答案】**$-\dfrac{1}{4}$

**【解析】**由 $$x\int_0^2 \mathrm{e}^{-(xt)^2}\mathrm{d}t\xlongequal{令xt=u}\int_0^{2x}\mathrm{e}^{-u^2}\mathrm{d}u,$$

得 $$f(x)=\int_0^{2x}\mathrm{e}^{-u^2}\mathrm{d}u+x^2,$$

于是$f'(x)=2\mathrm{e}^{-4x^2}+2x$，$f''(x)=-16x\mathrm{e}^{-4x^2}+2$，$f'(0)=2$，$f''(0)=2$，且$f(0)=0$，故

$$g''(0)=\frac{-f''(0)}{[f'(0)]^3}=-\frac{1}{4}.$$

**24.【解析】**当$x\to0^+$时，

$$F(x)=\int_0^x(\sin x-\sin t)\mathrm{d}t+\int_x^{\frac{\pi}{2}}(\sin t-\sin x)\mathrm{d}t$$

$$=x\sin x+\cos x-1+\cos x-\sin x\cdot\left(\frac{\pi}{2}-x\right)$$

$$=\left(2x-\frac{\pi}{2}\right)\sin x+2\cos x-1.$$

**方法一** 直接展开. $$\sin x=x+o(x^2),$$

$$\cos x=1-\frac{1}{2}x^2+o(x^2),$$

则 $$F(x)=\left(2x-\frac{\pi}{2}\right)\left[x+o(x^2)\right]+2\left[1-\frac{1}{2}x^2+o(x^2)\right]-1=1-\frac{\pi}{2}x+x^2+o(x^2),$$

因此$a=1$，$b=-\dfrac{\pi}{2}$，$c=1$.

配套课程公众号[研大]
联系微信19316378888

**方法二**　由 $F'(x)=2\sin x+\left(2x-\frac{\pi}{2}\right)\cos x-2\sin x=\left(2x-\frac{\pi}{2}\right)\cos x$，

$$F''(x)=2\cos x-\left(2x-\frac{\pi}{2}\right)\sin x,$$

故　$$F(0)=1,\ F'_+(0)=-\frac{\pi}{2},\ F''_+(0)=2,$$

$$F(x)=F(0)+F'_+(0)x+\frac{F''_+(0)}{2!}x^2+\cdots$$

$$=1-\frac{\pi}{2}x+x^2+\cdots,$$

即 $a=1$，$b=-\frac{\pi}{2}$，$c=1$.

**25.【解析】**当 $x\in(1,2)$ 时，

$$x\sqrt{3x^2-2x-1}=x\sqrt{(3x+1)(x-1)}=x(x-1)\sqrt{\frac{3x+1}{x-1}},$$

令 $t=\sqrt{\frac{3x+1}{x-1}}$，有 $\frac{3x+1}{x-1}=t^2$，解得 $x=\frac{t^2+1}{t^2-3}=1+\frac{4}{t^2-3}$，则 $\mathrm{d}x=-\frac{8t}{(t^2-3)^2}\mathrm{d}t$.

于是

$$\int_1^2\frac{\mathrm{d}x}{x\sqrt{3x^2-2x-1}}=\int_1^2\frac{1}{x(x-1)\sqrt{\frac{3x+1}{x-1}}}\mathrm{d}x$$

$$=\int_{+\infty}^{\sqrt{7}}\frac{1}{\frac{t^2+1}{t^2-3}\cdot\frac{4}{t^2-3}\cdot t}\cdot\frac{-8t}{(t^2-3)^2}\mathrm{d}t$$

$$=\int_{\sqrt{7}}^{+\infty}\frac{2}{1+t^2}\mathrm{d}t=2\arctan\ t\Big|_{\sqrt{7}}^{+\infty}=\pi-2\arctan\ \sqrt{7}.$$

**26.【答案】**(B)

**【解析】**原式 $=\int_{-1}^2\frac{\mathrm{d}x}{1+2^{\sqrt[3]{x-2}}}+\int_2^5\frac{\mathrm{d}x}{1+2^{\sqrt[3]{x-2}}}$，第一个积分中令 $x=4-t$，则

$$\int_{-1}^2\frac{\mathrm{d}x}{1+2^{\sqrt[3]{x-2}}}=\int_5^2\frac{-\mathrm{d}t}{1+2^{\sqrt[3]{2-t}}}=\int_2^5\frac{\mathrm{d}x}{1+2^{-\sqrt[3]{x-2}}},$$

则原式 $=\int_2^5\frac{\mathrm{d}x}{1+2^{-\sqrt[3]{x-2}}}+\int_2^5\frac{\mathrm{d}x}{1+2^{\sqrt[3]{x-2}}}=\int_2^5\left(\frac{1}{1+2^{-\sqrt[3]{x-2}}}+\frac{1}{1+2^{\sqrt[3]{x-2}}}\right)\mathrm{d}x=\int_2^5 1\mathrm{d}x=3$.

**27.【解析】**(1) $x^2y'+(x^2-3)y^2=0$ 整理得 $y'=\left(\frac{3}{x^2}-1\right)y^2$，即 $y^{-2}y'=\frac{3}{x^2}-1$，此为可分离变量型微

配套课程公众号[研大]
联系微信19316378888

分方程，即 $\frac{dy}{y^2}=\left(\frac{3}{x^2}-1\right)dx$，积分得 $-\frac{1}{y}=-\frac{3}{x}-x+C$，即 $y=\frac{x}{x^2-Cx+3}$. 由 $y(1)=1$，知 $\frac{1}{4-C}=1$，有 $C=3$，故

$$y(x)=\frac{x}{x^2-3x+3}.$$

(2) 由(1)知，

$$\int_0^3 y^2(x)dx=\int_0^3\frac{x^2}{(x^2-3x+3)^2}dx.$$

因

$$x^2-3x+3=\left(x-\frac{3}{2}\right)^2+\frac{3}{4},$$

令 $x-\frac{3}{2}=\frac{\sqrt{3}}{2}\tan u\left(|u|<\frac{\pi}{2}\right)$，则

$$(x^2-3x+3)^2=\left(\frac{3}{4}\sec^2 u\right)^2=\frac{9}{16}\sec^4 u,\ dx=\frac{\sqrt{3}}{2}\sec^2 u du.$$

当 $x=0$ 时，$u=-\frac{\pi}{3}$；当 $x=3$ 时，$u=\frac{\pi}{3}$. 于是

$$\int_0^3\frac{x^2}{(x^2-3x+3)^2}dx=\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}}\left(\frac{3}{4}\tan^2 u+\frac{3\sqrt{3}}{2}\tan u+\frac{9}{4}\right)\cdot\frac{16}{9}\cdot\frac{\sqrt{3}}{2}\cos^2 u du$$

$$=\frac{8}{3\sqrt{3}}\cdot 2\int_0^{\frac{\pi}{3}}\left(\frac{3}{4}\tan^2 u+\frac{9}{4}\right)\cos^2 u du$$

$$=\frac{4}{\sqrt{3}}\int_0^{\frac{\pi}{3}}(\sin^2 u+3\cos^2 u)du=\frac{4}{\sqrt{3}}\int_0^{\frac{\pi}{3}}(2+\cos 2u)du$$

$$=\frac{4}{\sqrt{3}}\left(2u+\frac{1}{2}\sin 2u\right)\Big|_0^{\frac{\pi}{3}}=\frac{8\sqrt{3}\pi}{9}+1.$$

**28.【答案】0**

**【解析】**

$$\int_0^{+\infty}\frac{x\ln x}{1+x^4}dx=\int_0^1\frac{x\ln x}{1+x^4}dx+\int_1^{+\infty}\frac{x\ln x}{1+x^4}dx\overset{记}{=\!=}I_1+I_2,$$

其中 $I_1$，$I_2$ 收敛.

而

$$I_1\overset{x=\frac{1}{t}}{=\!=}\int_{+\infty}^1\frac{\frac{1}{t}(-\ln t)}{1+\frac{1}{t^4}}\left(-\frac{1}{t^2}\right)dt=-\int_1^{+\infty}\frac{t\ln t}{1+t^4}dt=-I_2,$$

故原式 $=I_1+I_2=0$.

配套课程公众号[研大]
联系微信19316378888